MILITARY SPECIFICATION

TACTICAL 18.75 KB DIGITAL DATA REGENERATIVE REPEATER PORTABLE HYL-3( )/TSEC (UNITS OF)

1. SCOPE

1.1 This specification covers one type of tactical digital data regenerative repeater used to extend the range of FM/VHF Radio Sets AN/PRC-77( ) and AN/VRC-12( ) when used in the cipher mode either conditioned baseband or diphase with speech security equipments of the KY-8 family (KY-8 and KY-38). The designation is Tactical 18.75 KB Digital Data Regenerative Repeater Portable HYL-3( )/TSEC.

1.2 Appendix. - The appendix listed below forms a part of this specification:

RELIABILITY TEST PLAN FOR TACTICAL 18.75 KB DIGITAL DATA REGENERATIVE REPEATER, PORTABLE, HYL-3( )/TSEC (UNITS OF).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet Material Handling, wood, general construction required
QQ-S-781	Strapping, steel, flat
PPP-B-585	Boxes, Wood Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Folding, Fiberboard, Corrugated, Triple Wall

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SPECIFICATIONS (Contd)

FEDERAL

PPP-F-320	Fiberboard; Sheet, Stock and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and connectors)
PPP-T-76	Tape; Pressure-Sensitive, Adhesive, Paper, Water Resistance
PPP-T-97	Tape; Pressure-Sensitive, Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shock Test, HI (HIGH IMPACT); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection of Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized & Containerized Unit Loads 40" X 48" Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices

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STANDARDS (Contd)

MILITARY

MIL-STD-454	General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirement for
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement for
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

DL-SM-B-662146	HYL-3( )/TSEC
GL-SM-B-323379	HYL-3( )/TSEC Gage List
SC-GL-323115	Gage List for Battery Case CY-2562( )/PRC-25

3. REQUIREMENTS

3.1 Construction. - The equipment shall be constructed in accordance with the requirements of this specification, and Data List Drawing DL-SM-B-662146 including the requirements for parts, materials and processes therein.

3.2 First article samples. - The contractor shall furnish fifteen (15) First Article samples of equipment on order for approval, if required by the invitation for bids and the contract (see 4.3 and 6.2d).

3.3 Parts, materials and processes: general. - In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein, shall apply.

3.3.1 Printed wiring boards and printed wiring assemblies. - Materials and processes for printed wiring boards and printed wiring assemblies shall conform to MIL-P-11268.

3.4 Finish, protective. - The equipment shall be given a protective finish in accordance with MIL-F-14072 and the equipment drawing. This includes finish of hardware, such as handles, hinges, screws, etc. and necessary touch-up after mounting.

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3.5 Marking. -

3.5.1 General. - Marking shall conform to Specification MIL-M-13231 (see 4.4). Front panel marking shall be group I as described in that specification.

3.5.2 Visibility. - Wherever, practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. - Each Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC shall be serial numbered.

3.6 Preconditioning. -

3.6.1 Bounce preconditioning (see 4.6.1). - Each Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC produced shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (also see 4.5).

3.6.2 Electrical preconditioning (burn-in, or debugging) (see 4.6.2). - The contractor may elect to operate the First Article samples for a certain time prior to the start of First Article testing. Any such burn-in applied to the First Article samples shall be at no additional cost to the Government and must be applied to all items of equipment submitted for production acceptance. Failures occuring during this period may be considered non-relevant, but shall be recorded and reported. Burn-in time for all items of equipment submitted for production acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the First Article samples. Test levels and test conditions for burn-in shall be the same as specified in A30.2 for the production reliability acceptance test.

3.7 Service conditions. - The equipment shall meet the following service conditions.

3.7.1 Temperature. - The equipment shall be subjected to the temperature procedure as shown in Method 501 and 502 of MIL-STD-810.

3.7.1.1 Low (see 4.8.1.1). - During testing as specified in 4.8.1.1, the equipment shall meet the electrical performance requirements of 3.8 both during and after testing.

3.7.1.2 High (see 4.8.1.2). - During testing as specified in 4.8.1.2, the equipment shall meet the electrical performance requirements of 3.8 both during and after testing.

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3.7.2 Humidity (moisture-resistance) (see 4.8.2). - The equipment shall meet the electrical performance requirements of 3.8 both during and after testing as specified in 4.8.2.

3.7.3 Altitude (elevation) (see 4.8.3). - The equipment shall meet the electrical performance requirements of 3.8 both during and after testing as specified in 4.8.3.

3.7.4 Vibration, bounce and shock. -

3.7.4.1 Vibration internal (see 4.8.4). - The amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz. There shall be no physical damage.

3.7.4.2 Bounce and shock. - The equipment shall be capable of meeting the requirements of Table I.

TABLE I - BOUNCE AND SHOCK

Paragraph Heading	Paragraph	Performance After Test
Bounce (Vehicular)	4.8.5	Specified performance (NOTE A). No physical damage.
Bounce (loose cargo)	4.8.6	Specified performance (NOTE A). No physical damage.
Shock (Ballistic)	4.8.7	Specified performance (NOTE A). Any physical damage shall be minor only, as classified in MIL-STD-252

NOTE A: The equipment shall meet the electrical performance requirements of 3.8.

3.7.5 Shock (bench handling) (see 4.8.8). - The equipment shall meet the electrical performance requirements of 3.8 after testing as specified in 4.8.8.

3.7.6 Drop (see 4.8.9). - The equipment shall meet the electrical performance requirements of 3.8 after testing as specified in 4.8.9. Any physical damage shall be minor only, as classified in MIL-STD-252.

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3.7.7 Leakage (immersion) (see 4.8.10). - The equipment shall show no evidence of leakage after it is tested as specified in 4.8.10.

3.7.8 Dust (see 4.8.11). - The equipment shall meet the electrical performance requirements of 3.8 after testing as specified in 4.8.11.

3.7.9 Salt fog (see 4.8.12). - After testing as specified in 4.8.12, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.7.10 Fungus (see 4.8.13). - The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.8.13. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.7.11 Air seal (see 4.7). - When tested as specified in 4.7, the decrease in vacuum shall not exceed 0.01 pounds per square inch.

3.7.12 Operational inspection. - The operational test specified in 4.16 shall be performed to determine that the equipment is operable prior to packaging, and at such other times as required.

3.8 Electrical performance. -

3.8.1 Switching. -

3.8.1.1 Plain mode switching (see 4.12.4). - The HYL-3( )/TSEC shall pass traffic in the plain mode and automatically switch the radio equipment to the proper mode for retransmission.

3.8.1.2 Cipher (x mode) switching (see 4.12.5). - The HYL-3( )/TSEC shall pass traffic in cipher mode either conditioned baseband or conditioned diphase and automatically switch the radio equipment to the proper mode for retransmission.

3.8.2 Monitoring. -

3.8.2.1 Plain mode monitoring without speech security equipment (see 4.12.6). - The HYL-3( )/TSEC shall monitor traffic in the plain mode using a handset or other device.

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3.8.2.2 Plain mode monitoring with speech security equipment (see 4.12.7). - The HYL-3( )/TSEC shall monitor traffic in the plain mode with speech security equipment in the plain mode using a handset or other device.

3.8.2.3 Cipher mode monitoring (see 4.12.8). - The HYL-3( )/TSEC shall monitor traffic in the cipher mode with speech security equipment using a handset or other device.

3.8.3 Break-in. -

3.8.3.1 Plain mode break-in without speech security equipment (see 4.12.9). - The HYL-3( )/TSEC shall break-in in a selected direction while monitoring the other direction in the plain mode using a handset or other device.

3.8.3.2 Plain mode break-in with speech security equipment (see 4.12.10). - The HYL-3( )/TSEC shall break-in in a selected direction with speech security equipment in the plain mode, while monitoring the other direction using a handset or other device.

3.8.3.3 Cipher mode break-in (see 4.12.11). - The HYL-3( )/TSEC shall break-in in a selected direction using speech security equipment in the cipher mode while monitoring the other direction to detect the presence of the cipher using a handset or other device.

3.9 Electromagnetic interference. -

3.9.1 Electromagnetic compatibility (see 4.10). - The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461 for class IC. The following requirements of that standard are applicable:

CE 01	CE 04	CS 02	RE 02
CE 03	CS 01	CS 06	RS 03

3.10 Reliability. - The reliability of the equipment shall provide a specified (θ_o) mean-time-between-failure (MTBF) of 2000 hours. The required reliability shall be demonstrated in accordance with 4.11 of this specification

3.11 Interchangeability (see 4.9). - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.12 Technical literature (see 6.2). - Technical literature shall be furnished if specified in the contract.

3.13 Workmanship (see 4.13). - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and paragraph 3.1.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.14).

4.3 First article inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the item listed in paragraph 4.4, the preconditioning specified in paragraph 3.6, the inspections specified in group A, group B and group C (see Tables III, IV and V respectively), plus the reliability qualification test specified in paragraph 4.11.1. Each First Article sample shall be subjected to all the tests specified in group A and group B (see Tables III and IV respectively) prior to proceeding with reliability and environmental testing. Environmental testing shall be performed in the order and on the number of equipments specified in Table II. Reliability qualification testing shall be performed on ten (10) first article units in accordance with the applicable sections of Appendix A.

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4.3 First article inspection - (Contd) -

TABLE II - ORDER OF FIRST ARTICLE ENVIRONMENTAL INSPECTION

Inspection (Note 1)	Number of Samples				
	One	Two	Three	Four	Five
Non-damaging					
Temperature					
Low	1				1
High	2				2
Altitude (elevation)				1	
Leakage (immersion)		1 & 6	2 & 6		
Sand and dust			1		
Vibration (resonance search)		2	3		
EMI					3
Potentially damaging					
Humidity (moisture)	3				
Bounce, loose cargo		3			
Bounce, vehicular			4		
Shock, bench handling		4			
Shock, drop		5			
Shock, (ballistic)			5		
Fungus				2	
Salt atmosphere					4

NOTE 1: Other First Article inspection may precede, follow, or be interspersed between the non-damaging tests.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

ITEM	WHERE REQUIRED
Parts, Materials and Processes Used in Electronic Equipment	3.3
Finish	3.4
Marking	3.5
Workmanship	3.13

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4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to preconditioning (see 4.6) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6.1. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.4). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQLs indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government; except that the Air-Seal test (4.7) shall be next to last and the operational inspection (4.16) shall be last (see 6.5).

TABLE III - GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual-Mechanical	3.13	4.13		
Modules				
Z-ADX/TSEC, Power Converter			0.65%	2.5%
E-CAG/TSEC, Channel Selector and Signal Conditioner			0.65%	2.5%
E-CAF/TSEC, Output Driver and Crystal Oscillator			0.65%	2.5%
Power Converter + 12 volts A1			0.65%	2.5%
Power Converter + 12 volts input A2			0.65%	2.5%
Battery Box CY- ( )/U			0.65%	2.5%
Case, HYL-3( )			0.65%	2.5%
Complete HYL-3( ) Wired Chassis (less modules and case)			1.0%	4.0%
Assembled HYL-3( ) without case			1.0%	4.0%
Assembled HYL-3( ) with case			1.0%	4.0%

(Contd)

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4.5.1 Group A inspection - (Contd) -

TABLE III - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Electrical				
Plain mode switching	3.8.1.1	4.12.4		
X mode switching	3.8.1.2	4.12.5		
Plain mode monitoring without speech equipment	3.8.2.1	4.12.6	1.0% for the group	
Plain mode monitoring with speech equipment	3.8.2.2	4.12.7	All defects are considered major.	
Cipher mode monitoring	3.8.2.3	4.12.8		
Plain mode break-in without speech security equipment	3.8.3.1	4.12.9		
Plain mode break-in with speech security equipment	3.8.3.2	4.12.10		
X mode break-in	3.8.3.3	4.12.11		
Air seal test	3.7.11	4.7	1.0%	
Operational Inspection	3.7.12	4.16	1.0%	

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table IV and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on modules selected from end equipments that have been subjected to and met the Group A inspection. Inspection levels shall be S-4.

TABLE IV - GROUP B INSPECTION

Inspection	Req Para	Insp Para	AQL
Interchangeability			
Dimensional (Modules)	3.11	4.9.1	4.0%
Electrical (Modules)	3.11	4.9.2	4.0%

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4.5.3 Group C inspection. - This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

4.5.3.1 Sampling for inspection of equipment. - Units selected for Group C inspection shall be selected without regard to their quality in accordance with the following:

TABLE V - GROUP C INSPECTION

Inspection	Req Para	Insp Para
Subgroup 1		
Temperature		
Low	3.7.1.1	4.8.1.1
High	3.7.1.2	4.8.1.2
Vibration (internal)	3.7.4.1	4.8.4
Leakage (immersion)	3.7.7	4.8.10
Subgroup 2		
Humidity (moisture-resistance)	3.7.2	4.8.2
Bounce (vehicular)	3.7.4.2	4.8.5
Bounce (loose cargo)	3.7.4.2	4.8.6
Subgroup 3		
Shock ballistic	3.7.4.2	4.8.7
Shock drop	3.7.6	4.8.9
Shock (bench handling)	3.7.5	4.8.8
EMI	3.9	4.10
Altitude (elevation)	3.7.3	4.8.3
Subgroup 4		
Dust	3.7.8	4.8.11
Salt fog	3.7.9	4.8.12
Fungus	3.7.10	4.8.13

a. Subgroup 1. - For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every month. Both units shall be subjected to temperature, vibration and immersion testing in that order.

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4.5.3.1 Sampling for inspection of equipment - (Contd) -

b. Subgroup 2. - For this subgroup, three units shall be selected from the first production lot. For subsequent Group C inspection, three units shall be selected every two months. One unit shall be subjected to humidity testing, one unit shall be subjected to bounce (vehicular) testing, and one unit subjected to bounce (loose cargo) testing.

c. Subgroup 3. - For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every third month. All tests of this subgroup need not be performed on each unit. The testing may be divided between the two units as long as all subgroup tests are performed.

d. Subgroup 4. - For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every six months. The tests in this subgroup may be divided between the two units.

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable and (2) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative (QAR). After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspections may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Disposition of group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order after having met the requirements of 4.5.4.

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4.5.6 Group D inspection. - This inspection shall consist of the tests specified in Table VI, and shall be performed on units from lots which have been subjected to and met Group A inspection.

TABLE VI - GROUP D INSPECTION

Inspection	Req Para	Insp Para
Reliability	3.10	4.11.2

4.5.6.1 Noncompliance. - The contractor shall immediately investigate the cause of all failures in any lot in which a reject decision is reached and further report in writing within 48 hours to the procuring activity the results of the investigation and the proposed corrective action applicable to all production units which were manufactured under the same conditions which the Government considers subject to the same failures. The procuring activity shall have the final determination as to the nature of the failures; i.e., whether the failures are random in nature (no corrective action required) or subject to pattern failures. Acceptance by the Government of all in-house items shall be withheld until the contractor's proposed corrective actions have been accepted by the procuring activity and implemented as directed by the Government on all contract items in-house. A retest shall be imposed if it is deemed necessary by the Government to demonstrate the effectiveness of the corrective actions and the conditions of 4.11.2 shall then apply.

4.6 Preconditioning. -

4.6.1 Bounce preconditioning (see 3.6.1). - The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to quality conformance inspection.

4.6.2 Electrical preconditioning (burn-in debugging) (see 3.6.2). - If the contractor elects to perform electrical preconditioning, each unit shall be subjected to the test conditions cited in A30.4 and shall meet the requirements of 3.6.2.

4.7 Air seal test (see 3.7.11). - The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure

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4.7 Air seal test (see 3.7.11) - (Contd) - surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment inclosure when no transit case is provided. The vacuum shall then be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period, the decrease in vacuum shall not exceed 0.01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily. Battery Box CY- ( )/U shall be removed for this test.

4.8 Service conditions tests. - The following tests shall be performed on the equipment (see 3.7):

4.8.1 Temperature test. -

4.8.1.1 Low temperature test (see 3.7.1.1). - The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -65°F, and the operating temperature shall be -31°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8. Maintain storage temperature for 4 hours after temperature stabilization.

4.8.1.2 High temperature test (see 3.7.1.2). - The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating and non-operating (STORAGE) temperature shall be 155°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.

4.8.2 Humidity test (moisture-resistance) test (see 3.7.2). - The equipment shall be tested in accordance with Method 507, Procedure III, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8. Measurements shall be made during the periods shown on the figure for the cycle. Do not move the equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and then the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.8.3 Altitude test (elevation) test (see 3.7.3). - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD 810. Prior to testing, the equipment shall meet the full specification performance for the measurements specified in 3.8.

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4.8.4 Vibration internal test (see 3.7.4.1). - The equipment shall be tested in accordance with Method 514.1, Procedure IX, Part 1 of MIL-STD-810.

4.8.5 Bounce vehicular test (see 3.7.4.2). - The equipment shall be tested in accordance with Method 514.1, Procedure IX, Part 2 of MIL-STD-810. The HYL-3/TSEC Digital Data Regenerative Repeater shall be tested while mounted on the Government-furnished Shock Mount MT-3823( )/GRC. The equipment with its shock mount shall be secured to the vehicular adapter plate of the shock machine, using holes provided for vehicular installation. The adapter plate shall be free of cracks and broken welds, and all unused holes, except the 2-inch breather holes, shall be plugged. The test sample shall be mounted on the vehicular adapter plate with the test sample centered. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

4.8.6 Bounce test (loose cargo) (see 3.7.4.2). - The equipment shall be tested in accordance with Method 514.1, Procedure XI, Part 2 of MIL-STD-810.

4.8.7 Shock test (ballistic) (see 3.7.4.2). - The test shall be conducted on the "Shock Testing Machine for Lightweight Equipment" shown in Figure 1 of Specification MIL-S-901. The HYL-3 to be tested shall be mounted on a Government-furnished Shock Mount MT-3823( )/GRC. The equipment with its Shock Mount, shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows; one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternate to reorienting the plate for the blows on the side of the plate, an equivalent rotation of the equipment under test is permissible.

4.8.8 Shock test (bench handling) (see 3.7.5). - The equipment shall be tested in accordance with Method 516.1, Procedure V, of MIL-STD-810.

4.8.9 Drop test (see 3.7.6). - The equipment shall be tested in accordance with Method 516.1, Procedure II, of MIL-STD-810.

4.8.10 Leakage test (immersion) (see 3.7.7). - The equipment shall be tested in accordance with Method 512, Procedure I, of MIL-STD-810.

4.8.11 Dust test (see 3.7.8). - The equipment shall be tested in accordance with Method 510, Procedure I, of MIL-STD-810, except that the equipment shall be operating for one hour of the 6 hour periods of steps 1 and 3.

4.8.12 Salt fog test (see 3.7.9). - The equipment shall be subjected to the test of Method 509, Procedure I, of MIL-STD-810. Operation of the equipment shall not be required and the failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surface that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces.

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4.8.13 Fungus test (see 3.7.10). - The equipment shall be subjected to the test of Method 508, Procedure I, of MIL-STD-810. For gasket sealed or other non-hermetic sealed equipments, the parts and materials of the chassis shall be sprayed with spore suspension. The interior and exterior of the case shall also be sprayed. The equipment shall remain in an open condition during the test. Operation is not required.

4.9 Inspection for interchangeability (see 3.11). - The following dimensions and performance shall be gaged or measured to determine compliance with the interchangeability requirements of 3.11.

4.9.1 Mechanical (dimensions). - Gage points covered by the following gage lists shall be measured: GL-SM-B-323379 (HYL-3( )/TSEC) and SC-GL-323115 (Battery Box CY-2562( )/PRC-25).

4.9.2 Electrical performance (modules). - Modules shall be tested to determine compliance with the performance requirements specified in the applicable test procedure drawing referenced in drawings covering construction of the modules.

4.10 Electromagnetic compatibility (see 3.9.1). - The equipment shall be tested for compliance with the requirements of Standard MIL-STD-461. The test setup and procedures employed shall conform with the measurement techniques of Standard MIL-STD-462.

4.11 Reliability (see 3.10). - The following reliability tests shall be performed on the equipment:

4.11.1 Reliability qualification test. - The Reliability Qualification Test shall be performed as part of First Article testing. The reliability of the equipment shall be demonstrated in accordance with the Reliability Qualification Test procedure specified in Appendix A of this specification to determine compliance with 3.10.

4.11.2 Production reliability acceptance test. - The reliability of the production equipment shall be demonstrated in accordance with the Production Reliability Acceptance Test procedure specified in Appendix A of this specification to determine compliance with 3.10.

4.12 Electrical tests. -

4.12.1 Standard conditions. - Unless otherwise specified, all tests shall be performed with a nominal 12.5 vdc power supply or Battery BA-4386( )/PRC-25, and shall be conducted at room ambient temperature, pressure and humidity.

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4.12.2 Special test equipment. - Unless otherwise specified, special test equipment shall be provided by the contractor subject to approval of the Government.

a. Test Adapter Box (see Fig. 3A and 3B).

b. Signal Generator, Special Purpose For X-Mode Signal Simulation (see Fig. 4).

4.12.3 Performance tests. - All tests are conducted with the Test Adapter Box (Figure 3A and 3B) and special purpose signal generator (Figure 4). Interconnection of cables and test equipment shall be in accordance with Figure 1.

4.12.4 Plain mode switching (see 3.8.1.1). - Set the HYL-3 controls to the following positions:

FUNCTION	RETRANS
POWER	ON

4.12.4.1 Signal from R/T-1. -

a. Place the R/T-1 RETRANS switch to the RETRANS (closed) position. Observe and record the following conditions of the R/T-2 test adapter lamps:

TONE DISABLE	OFF
PTT	ON

b. Apply the audio signal generator to the R/T-1 Audio out test point. Adjust for a 1.0 KHz audio input amplitude of 1.0 (± 0.02) Vrms. With the audio VTVM, measure and record this test point voltage.

c. With the audio VTVM, measure and record the 1.0 KHz signal which appears on the R/T-2 AUDIO IN test point. The signal should be 46 (± 3.0) db down from what was measured in step b above.

d. With the audio VTVM, measure and record the 1.0 KHz signal which appears at the R/T-2 AUX T.P. The signal should be 0 (+0, -.5) db below that measured in b above.

e. Place the RETRANS switch to OFF (open) and remove the audio generator and VTVM.

4.12.4.2 Signal from R/T-2. - Repeat paragraph 4.12.4.1a, b, c, d and e except that R/T-2 and R/T-1 are interchanged.

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4.12.5 Cipher (X-mode) switching (see 3.8.1.2). - Set the HYL-3( )/TSEC controls to the following positions:

FUNCTION	RETRANS
POWER	ON
VOLUME	MAXIMUM CW

a. Place the X mode PLAIN/X MODE switch to the X MODE position for all X MODE tests.

b. With no input signal to the HYL-3( )/TSEC, observe and record the following conditions of the R/T-1 and R/T-2 test adapter lamps:

TONE DISABLE	ON
PTT	OFF

4.12.5.1 Conditioned baseband signal from R/T-1. -

a. Apply the conditioned baseband signal from the X mode signal generator to the R/T-1 X mode out test point. With the oscilloscope, measure and record the peak to peak amplitude of this signal. The amplitude shall be 12.0 V. (+0, -1.0V.). Refer to Figure 2A for pulse measurements. With the oscilloscope, measure and record the pulse width (tw). It shall be 53.3 (± 4) usec. Measure and record the pulse period (tp). It shall be 426.4 (± 16) uSEC.

b. Place the test adapter diphase/baseband switch to baseband.

c. Observe and record the following conditions of the R/T-2 test adapter lamps:

TONE DISABLE	ON
PTT	ON

d. With the oscilloscope, monitor the R/T-2 X MODE IN test point. The signal at this point should be similar to the inverted output or the non-inverted output of Figure 2A. Sufficient time should be allowed after application of the input signal to allow for circuit acquisition and lock on to the input signal. After acquisition, monitor this test point for approximately 30 seconds to ensure that no switching between the inverted and non-inverted waveforms occurs. No switching indicates that no errors have been introduced into the signal by the HYL-3/TSEC circuits. The peak to peak amplitude of this signal should be 4.6 volts (± .7 volts). Measure and record delay time of this signal from the input signal (tDPP or tDPN as the case may be). The time delay shall be 100 (± 20) uSEC.

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4.12.5.2 Conditioned baseband signal from R/T-2. - Repeat 4.12.5.1a, b, and c except that R/T-1 and R/T-2 functions are interchanged.

4.12.5.3 Conditioned diphase signal from R/T-1. -

a. Place the test adapter diphase/baseband switch to diphase.

b. Apply the conditioned diphase signal from the X-mode signal generator to the R/T-1 X-mode out test point. (Refer to Figure 3A). Adjust this signal level to 1.0 volts (± .5 volts) by means of R1 (Figure 4). Repeat 4.12.5.1d and 4.12.5.2. The peak to peak amplitude of the output signal shall be 8.5 volts (± 1.2 volts) (see Figure 2B).

4.12.5.4 Conditioned diphase signal from R/T-2. - Repeat 4.12.5.3 a and b except that R/T-1 and R/T-2 functions are interchanged.

4.12.6 Plain mode monitoring without speech security equipment (see 3.8.2.1). - Place the HYL-3( )/TSEC controls to the following positions:

FUNCTION	RETRANS
POWER	ON
VOLUME	MAXIMUM CW

4.12.6.1 Signal from R/T-1. -

a. Place the R/T-1 RETRANS switch to the RETRANS (closed) position.

b. Apply the audio signal generator to the R/T-1 audio out test point. Adjust for a 1.0 KHz audio input amplitude of 1.0 (± 0.02) Vrms. With the audio VTVM, measure and record this voltage. With the distortion analyzer, measure and record the harmonic distortion of this signal. It shall be less than 1%.

c. With the audio VTVM, measure and record the 1.0 KHz signal which appears on the handset audio in test point. It should be 1.0 (± 0.1) Vrms. Measure and record the harmonic distortion of this signal. It shall be less than 5%.

d. Adjust the volume control to its approximate center position. With the audio VTVM, measure the HYL-3( )/TSEC handset audio in test point. This measurement should be approximately 20 db down from the measurement in part c above.

e. Place the RETRANS switch in the OFF position (open) and remove the audio generator and VTVM from the test points.

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4.12.7 Plain mode monitoring with speech security equipment (see 3.8.2.2). - Place the HYL-3( )/TSEC controls to the following positions:

FUNCTION	RETRANS
POWER	ON
VOLUME	MAXIMUM CW

4.12.7.1 Signal from R/T-1. -

a. Place the R/T-1 RETRANS switch to the RETRANS position (closed).

b. Apply the audio signal generator to the R/T-1 AUDIO OUT. Adjust for a 1.0 KHz audio input amplitude of 1.0 (+ 0.02) Vrms. With the audio VTVM, measure and record this voltage. With the distortion analyzer, measure and record the harmonic distortion of this signal. It shall be less than 1%.

c. With the audio VTVM, measure and record the 1.0 KHz signal which appears on the X mode MONITOR IN test point. It shall be 1.0 (+ 0.15) Vrms. Measure and record the distortion of this signal. It shall be less than 5%.

4.12.8 Cipher mode monitoring (see 3.8.2.3). - Place the HYL-3( )/TSEC controls to the following positions:

FUNCTION	RETRANS
POWER	ON
VOLUME	MAXIMUM CW

4.12.8.1 Signal from R/T-1. -

a. Apply the conditioned baseband signal from the X-mode signal generator to the R/T-1 X-mode out test point. Observe and record the following conditions of the R/T-2 test adapter lamps:

TONE DISABLE	ON
PTT	ON

b. With the oscilloscope, monitor the x-mode digital in (Fig. 3B) test point. Measure and record the peak to peak amplitude. It shall be 12 (± 1.5) volts peak to peak.

c. With the oscilloscope, monitor the x-mode unit monitor in (Fig. 3B) test point. Observe that this is an inverted form of the digital signal appearing on the digital in test point. Measure and record this output. It shall be 1.0 volts (± 0.3) volts.

4.12.9 Plain mode break-in without speech security equipment (see 3.8.3.1). -

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4.12.9.1 Break-in R/T-1. - Place the HYL-3( )/TSEC controls in the following positions:

FUNCTION	XMIT 1
POWER	ON
VOLUME	MAXIMUM CW

a. Apply a 1.0 KHz signal from the audio generator to the headset mike test point (Fig. 3B). Adjust the amplitude to 1.0 (± 0.02) Vrms. Measure and record. Measure and record the harmonic distortion of this signal. It shall be less than 1%.

b. Connect the audio VTVM to the R/T-1 AUDIO IN test point. The 1 KHz signal shall have an amplitude of 1.0 Vrms ± 5%. Measure and record.

c. Observe and record the following conditions of the R/T-1 lamps:

TONE DISABLE	ON
PTT	OFF

d. Place the handset PTT switch to the ON position and observe and record the following conditions of the R/T-1 lamps:

TONE DISABLE	OFF
PTT	ON

e. Place the handset PTT switch to the OFF position.

4.12.9.2 Break-in R/T-2. - Place the HYL-3( )/TSEC controls in the following positions:

FUNCTION	XMIT 2
POWER	ON

Repeat 4.12.9.1a, b, c, d and e, except that R/T-2 and R/T-1 are interchanged.

4.12.10 Plain mode break-in with speech security equipment (see 3.8.3.2). -

4.12.10.1 Break-in R/T-2. - Place the HYL-3( )/TSEC controls in the following positions:

FUNCTION	XMIT 2
POWER	ON
VOLUME	MAXIMUM CW

Verify that the Plain/X Mode switch is in the Plain position. Confirm that the X-Mode PTT switch is off.

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4.12.10.1 Break-in R/T-2.- (Contd) -

a. Apply a 1.0 KHz signal from the audio generator to the X mode MIKE. Adjust the amplitude to 1.0 (± 0.02) Vrms. Measure and record. Measure and record the harmonic distortion of this signal. It shall be less than 1%.

b. Connect the audio VTVM to the R/T-2 AUDIO IN. The 1 KHz signal shall have an amplitude of 1.0 Vrms ± 5%. Measure and record.

c. Observe and record the following conditions of the R/T-2 test adapter lamps:

TONE DISABLE	ON
PTT	OFF

d. Place the X mode PTT switch to the ON position and observe and record the following conditions of the R/T-2 lamps:

TONE DISABLE	OFF
PTT	ON

e. Place the X mode PTT switch to the off position.

4.12.10.2 Break-in R/T-1. - Place the HYL-3( )/TSEC controls in the following positions:

FUNCTION	XMIT 1
POWER	ON

Repeat 4.12.10.1a, b, c, and d except that R/T-1 and R/T-2 are interchanged.

4.12.11 X mode break-in (see 3.8.3.3). -

4.12.11.1 Break-in R/T-1. - Place the HYL-3( )/TSEC controls in the following positions:

FUNCTION	XMIT 1
POWER	ON
VOLUME	MAXIMUM CW

a. Verify that the Plain/X mode switch is in the X MODE position. Place the X mode PTT switch to the ON position. Observe and record the following conditions of the test lamps:

R/T-1 TONE DISABLE	ON
R/T-1 PTT	ON
R/T-2 TONE DISABLE	ON
R/T-2 PTT	OFF

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4.12.11.1 Break-in R/T-1 - (Contd) -

b. Apply the conditioned baseband signal from the X mode signal generator to the X mode DIGITAL OUT test point. With the oscilloscope, measure and record the peak to peak amplitude of this signal. It shall be 1.0 (+1.0, -0.5) volts.

c. Place test adapter Diphase/Baseband switch to baseband.

d. With the oscilloscope, monitor the R/T-1 X MODE IN test point and verify the X mode signal generator signal is present. Measure and record the amplitude. It shall be 4.6 volts (± .7 v).

4.12.11.2 Break-in R/T-2. - Place the HYL-3( )/TSEC controls in the following positions:

FUNCTION	XMIT 2
POWER	ON

a. Verify that the PLAIN/X MODE switch is in the X MODE position. Place the X MODE PTT switch to the ON position. Observe and record the following conditions of the test adapter lamps:

R/T-1 TONE DISABLE	ON
R/T-1 PTT	OFF
R/T-2 TONE DISABLE	ON
R/T-2 PTT	ON

b. Apply the conditioned baseband signal from the X mode signal generator to the X mode Digital Out. With the oscilloscope, measure and record the peak to peak amplitude of this signal. It shall be 1.0 (+1.0, -0.5) volts.

c. Place test adapter Diphase/Baseband switch to baseband.

d. With the oscilloscope, monitor the R/T-2 X MODE IN and verify that the X mode signal is present. Measure and record the amplitude. It shall be 4.6 volts (± .7 v).

4.13 Visual and mechanical inspection (see 3.13). - Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.14 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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4.15 Rough handling test (preparation for delivery). - When rough handling test in accordance with MIL-P-116 is required by the contract (see 6.2), the electrical operation performance tests (4.16) shall be conducted to determine freedom from operational malfunction caused by rough handling.

4.16 Operational inspection. - All equipments to be operationally inspected shall be tested in accordance with 4.12.5, 4.12.5.1 and 4.12.6.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see 6.2c).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as prescribed herein.

5.1.1.4.1 Technical literature. - Each Operators Manual (TM 11-5810-247-10) shall be packaged method IC-1.

5.1.1.4.2 Cables. - Each Cable CX-12066 shall be packaged in accordance with MIL-C-55442.

5.1.1.4.3 Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC. - Each Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC shall be individually packaged method III as follows: Cushion the unit on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, variety SW, grade W5c, class weather resistant, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item with the cables packaged as specified in 5.1.1.4.2 within a close fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

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5.1.2 Level C. - Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents (see 6.2c).

5.2.1 Level A. -

5.2.1.1 Palletized load. - A quantity of Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC, packaged as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the item, packages as specified in 5.1, shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2 or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the gross weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, class B.

5.2.2 Level B. -

5.2.2.1 Palletized load. - A quantity of Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Digital Data Regenerative Repeater, Portable

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5.2.2.2 Less than palletized load - (Contd) - HYL-3( )/TSEC, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather resistant, variety DW. The gross weight of the boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C. -

5.2.3.1 Palletized load. - A quantity of Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The Tactical 18.75 KB Digital Data Regenerative Repeater, Portable HYL-3( )/TSEC is a rugged, lightweight, compact equipment which shall extend the range of FM/VHF Radio Sets AN/PRC-77( ) and AN/VRC-12( ) when used in the cipher mode either conditioned baseband or diphase with speech security equipments of the KY-8 family (KY-8, and KY-38).

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Type required (see 1.1).

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6.2 Ordering data - (Contd) -

c. Levels of preservation, packaging and packing required for shipment (Level A, Level B, or Level C) (see 5.1).

d. First Article required (see 3.2).

e. Marking and shipping of samples.

f. Technical literature required (see 3.12).

g. If the rough handling and functional tests are required (see 4.15).

6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example HYL-3W/TSEC. As soon as possible after the award of contract, the contractor should apply to the Contracting Officer for nomenclature in accordance with the applicable clause of the contract.

6.4 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Location of air-seal and operational tests. - It is desirable that the air-seal test (4.7) and operational test (4.16) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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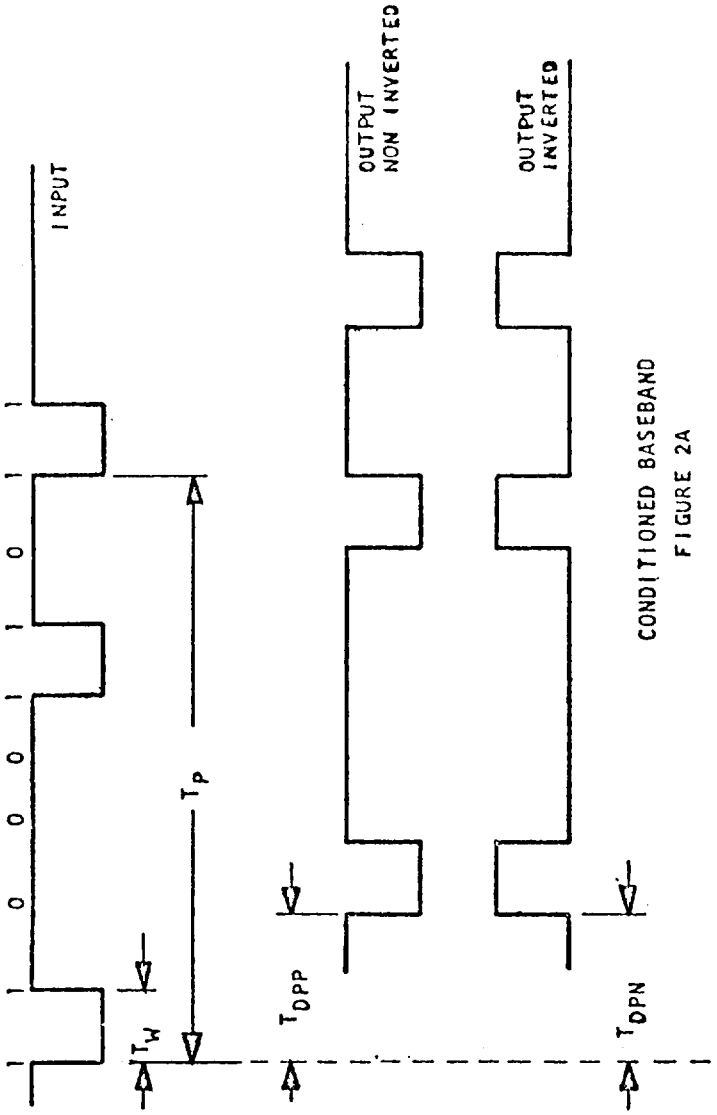
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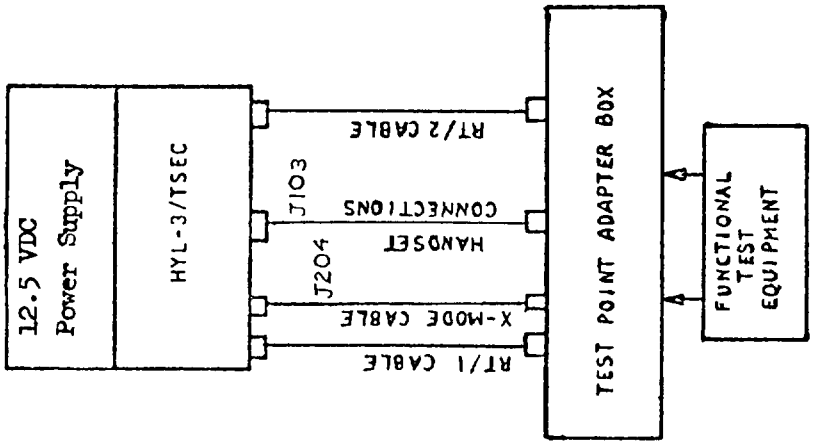
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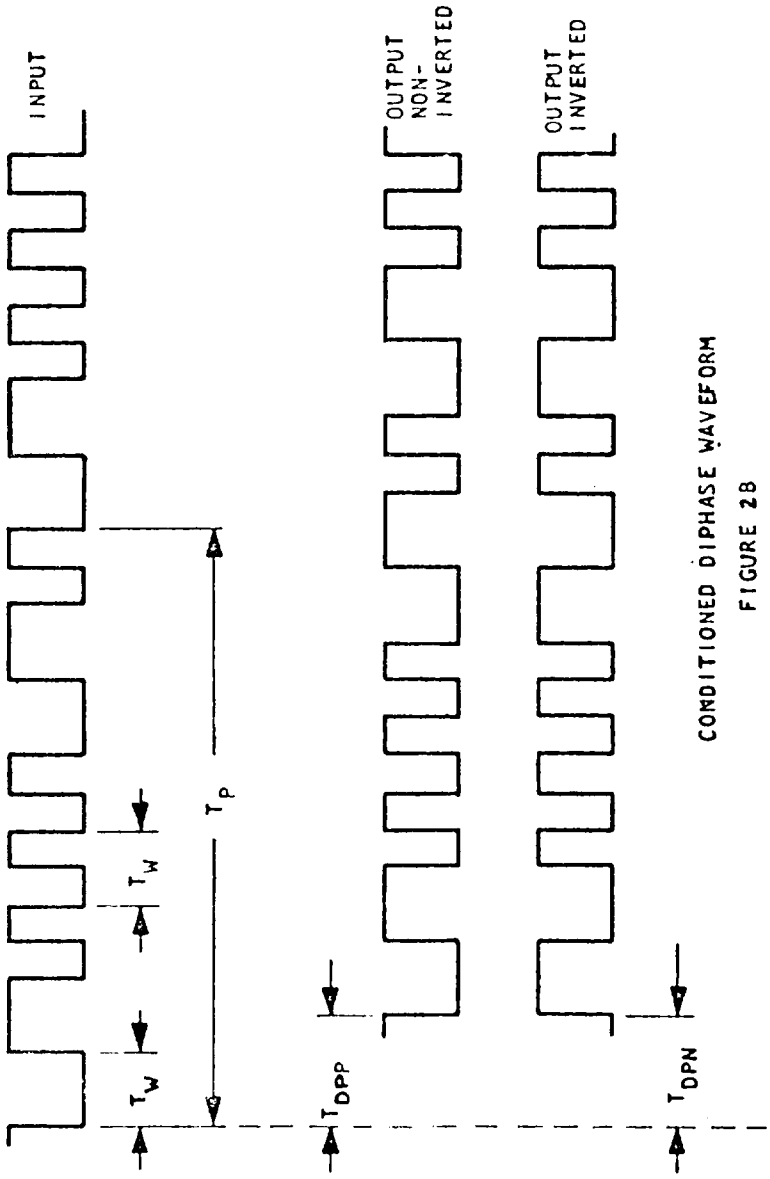


CONDITIONED BASEBAND
FIGURE 2A



CABLE INTERCONNECTION DIAGRAM
FIGURE 1

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CONDITIONED DIPHASE WAVEFORM
FIGURE 2B

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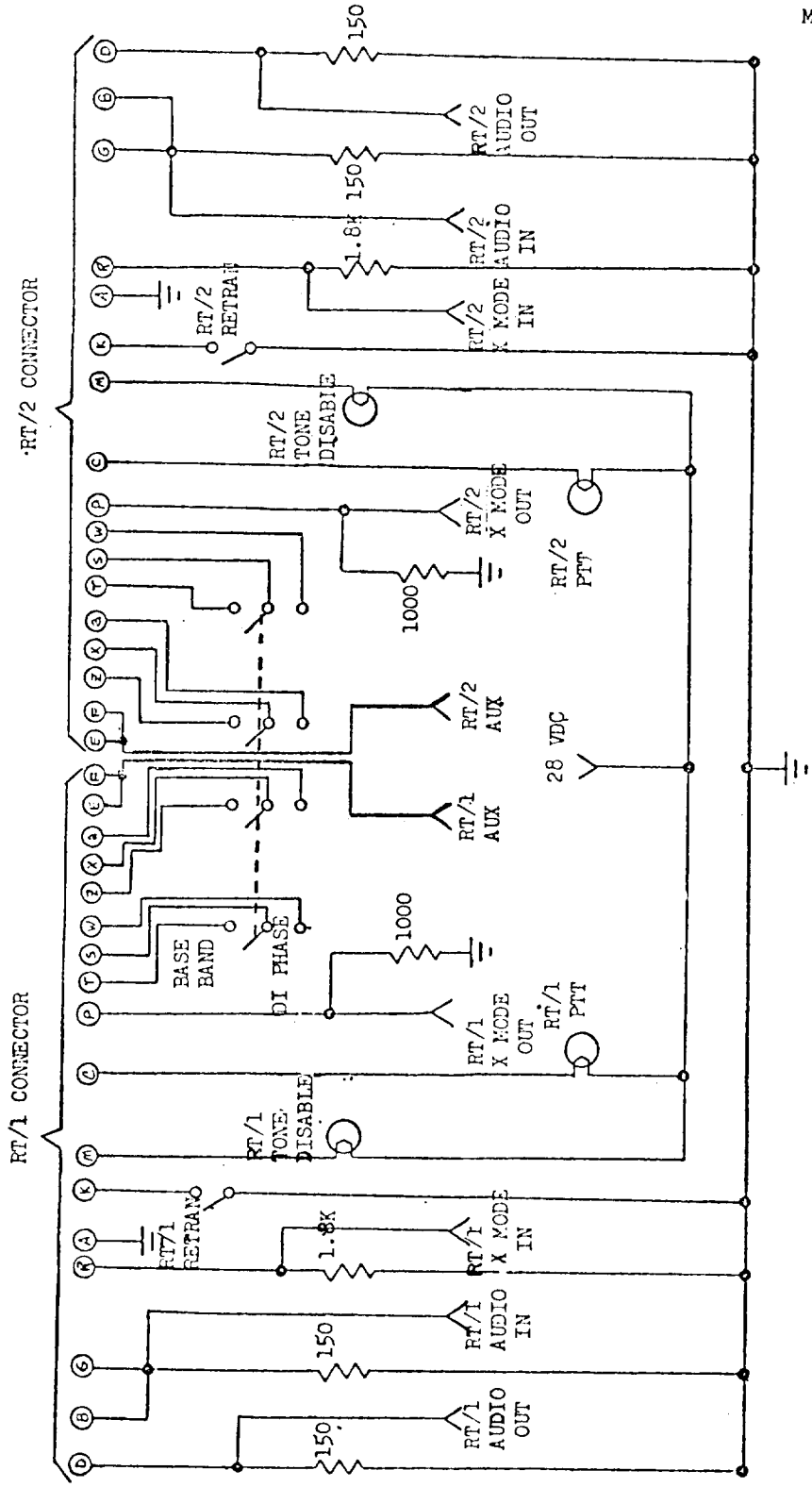
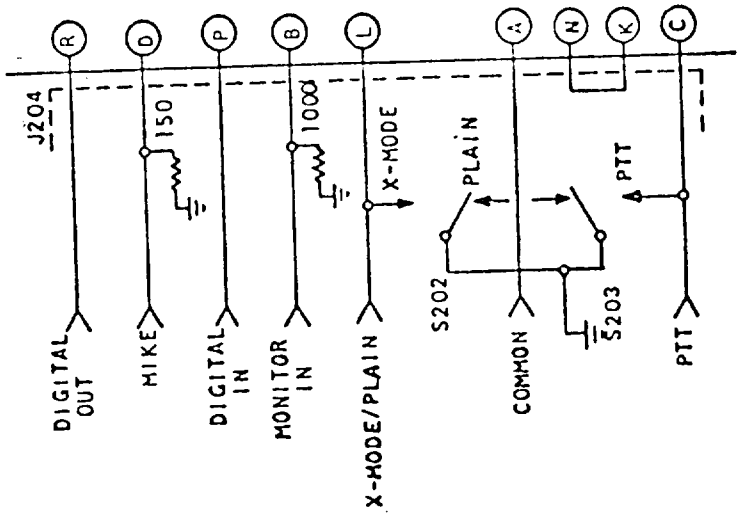


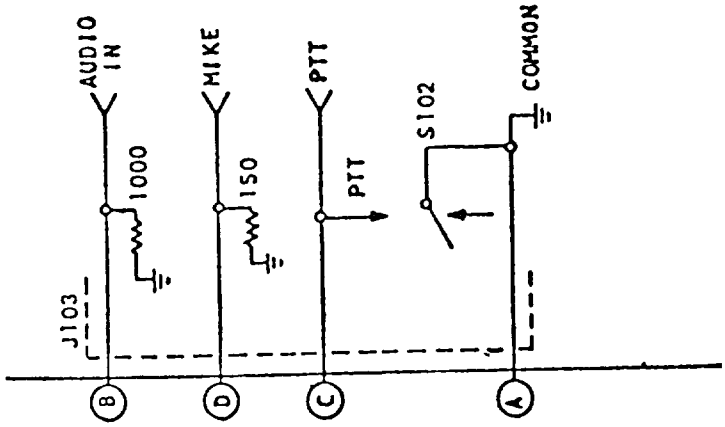
FIG. 3A HYL-3/TSEC TEST ADAPTER BOX

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X-MODE ADAPTER BOX DIAGRAM

TEST ADAPTER BOX
FIG 3B



HANDSET TEST ADAPTER DIAGRAM

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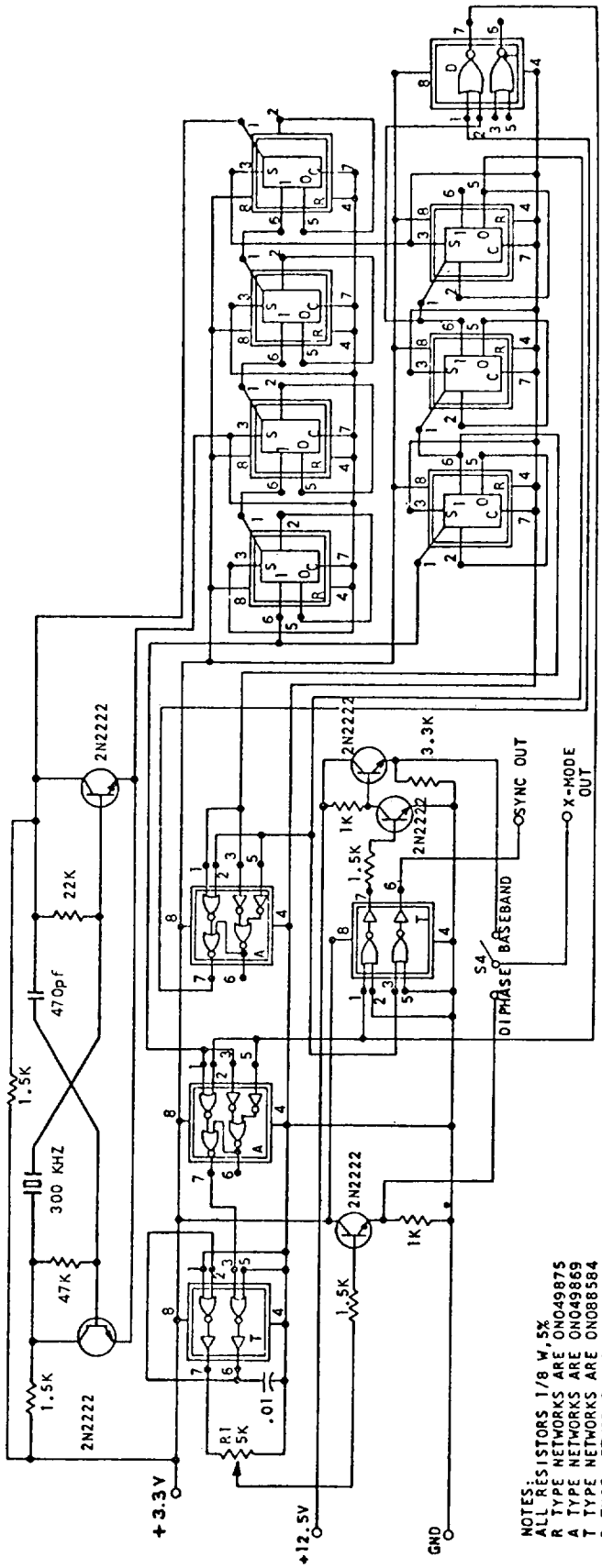


FIGURE 4
X-MODE GENERATOR

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APPENDIX A

RELIABILITY TEST PLAN FOR TACTICAL 18.75 KB DIGITAL DATA REGENERATIVE REPEATER, PORTABLE, HYL-3( )/TSEC (UNITS OF)

A10.0 SCOPE. - This plan defines the methods and procedures to be used to perform the Reliability Qualification Test and the Production Reliability Acceptance Test.

A20.0 APPLICABLE DOCUMENTS. - The following publications form a part of this document to the extent referenced herein: MIL-STD-781 Reliability Tests Exponential Distribution.

A30.0 RELIABILITY REQUIREMENTS. -

A30.1 Reliability qualification test. - The contractor shall perform this test as part of First Article testing. The reliability of the equipment shall be demonstrated by selecting ten (10) First Article samples that have passed Group A and Group B inspection and subjecting them to the requirements of Test Plan IV of MIL-STD-781. The equipment shall demonstrate a specified MTBF of 2000 hours. Test levels and test conditions shall be as specified in paragraph A30.4. No single equipment "on" time shall be less than one half the average operating time of all equipments on test.

A30.2 Production reliability acceptance test. - The reliability of production equipment shall be demonstrated in accordance with Figure A1 (derived from reject line of Test Plan IV, MIL-STD-781). Each unit of each lot of equipment that has been subjected to and met Group A inspection shall be tested for a duration of 200 hours under the test conditions and test levels specified in A30.4. The equipment shall demonstrate a specified MTBF of 2000 hours.

A30.2.1 Evaluation of test. - To determine whether the MTBF is being met at any time during the contract, the operation test hours and the failures (not counting burn-in failures or burn-in operation time) shall be totaled and the results compared with the reject line of Figure A1. These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. If the line of plotted test results indicates a reject situation, that is, if the plotted point crosses the reject line, then the procuring activity shall be notified immediately and corrective action in accordance with 4.5.6.1 shall be instituted.

A30.3 Electrical preconditioning (burn-in, or debugging). - Electrical preconditioning shall be as specified in 3.6.2.

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A30.4 Test conditions. -

A30.4.1 Cycle. - The testing shall be conducted on a 24 hour, 7-day-per-week basis as shown in Figure A2.

A30.4.2 Duty cycle. - Power shall be applied to all units under test continuously in order to expedite completion of the test.

A30.4.3 Environmental and operation stresses. - Vibration and Temperature cycling requirements shall be as specified in Test Level E of MIL-STD-781 except the low temperature test level shall be -31°F. Equipment ON-OFF cycling and Input Voltage cycling are not applicable. Temperature and vibration cycling shall be as shown in Figure A2.

A30.4.4 Performance parameters to be measured. - Each HYL-3 shall meet all the electrical requirements specified in paragraph 3.8 at low temperature, high temperature and room temperature. Testing shall be performed at each of these temperatures on a once a day basis.

A30.5 The following requirements apply to both the Reliability Qualification Test and the Production Reliability Acceptance Test:

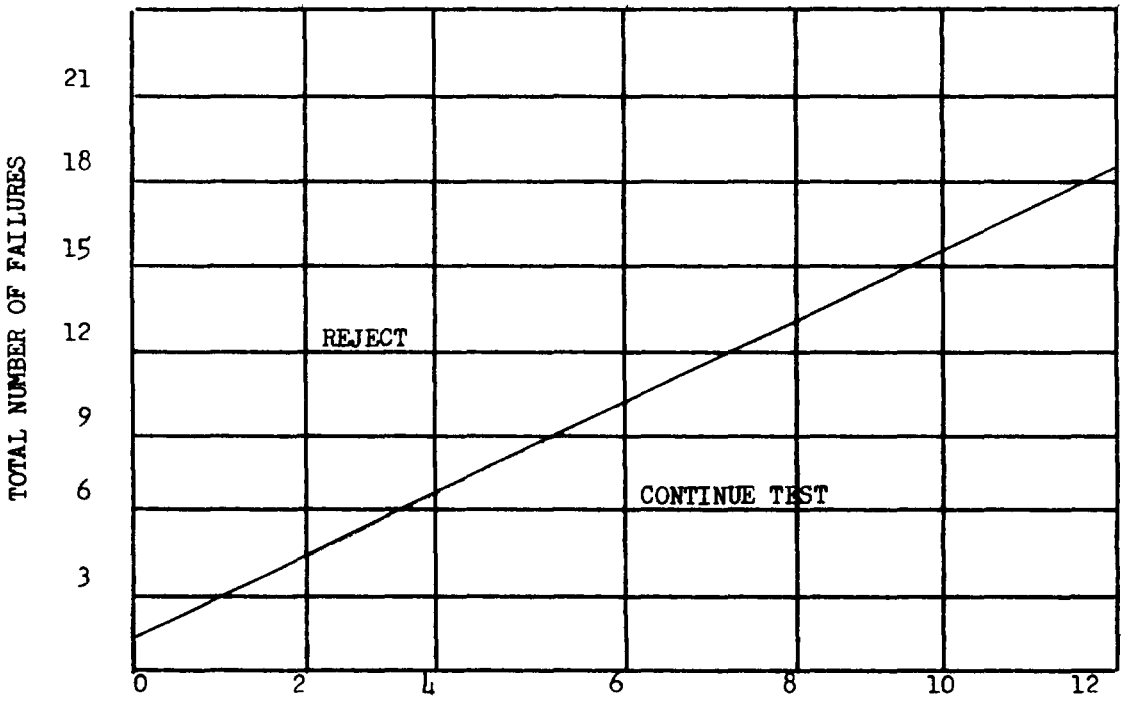
A30.5.1 Failure actions, categories and confirmation. - Failure Actions, Categories and Confirmation shall be as specified in MIL-STD-781. All failures occurring on the Government Furnished Equipment (i.e., E-BOM and E-BOL printed wiring assemblies) shall be classed as non-relevant and not counted in determining the success or failure of the equipments to pass the test.

A30.5.2 Verifying repair. - Post repair verification shall be as specified in MIL-STD-781.

A30.5.3 Preventive maintenance. - No preventive maintenance shall be performed on the HYL-3( ) during the Reliability Test. However, operator controls shall be adjusted as required.

A30.5.4 Test reports and failure reporting and analysis. - Test reports, failure reporting and failure analysis shall be as specified in the contract.

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No. of Failures	Total Test Time * Reject (Equal or less)	No. of Failures	Total Test Time * Reject (Equal or less)	No. of Failures	Total Test Time * Reject (Equal or less)
0	N/A	14	8.71	28	18.51
1	N/A	15	9.42	29	19.21
2	.35	16	10.10	30	19.91
3	1.04	17	10.81	31	20.62
4	1.73	18	11.50	32	21.31
5	2.43	19	12.21	33	22.01
6	3.12	20	12.92	34	22.72
7	3.81	21	13.61	35	23.42
8	4.51	22	14.30	36	24.10
9	5.21	23	14.99	37	24.81
10	5.91	24	15.71	38	25.50
11	6.60	25	16.40	39	26.21
12	7.32	26	17.11	40	26.90
13	8.02	27	17.82	41	27.60

* Total test time is total hours of "equipment on" time and is expressed in multiples of the specified MTBF.

FIGURE A1 ACCEPT-REJECT CRITERIA

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SPECIFICATION ANALYSIS SHEET

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SPECIFICATION MIL-C-55346(EL) TACTICAL 18.75 KB DIGITAL DATA REGENERATIVE REPEATER PORTABLE HYL-3( )/TSEC (UNITS OF)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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